# FabriCAD

Status report — Thursday 22 October 2026, 10:00

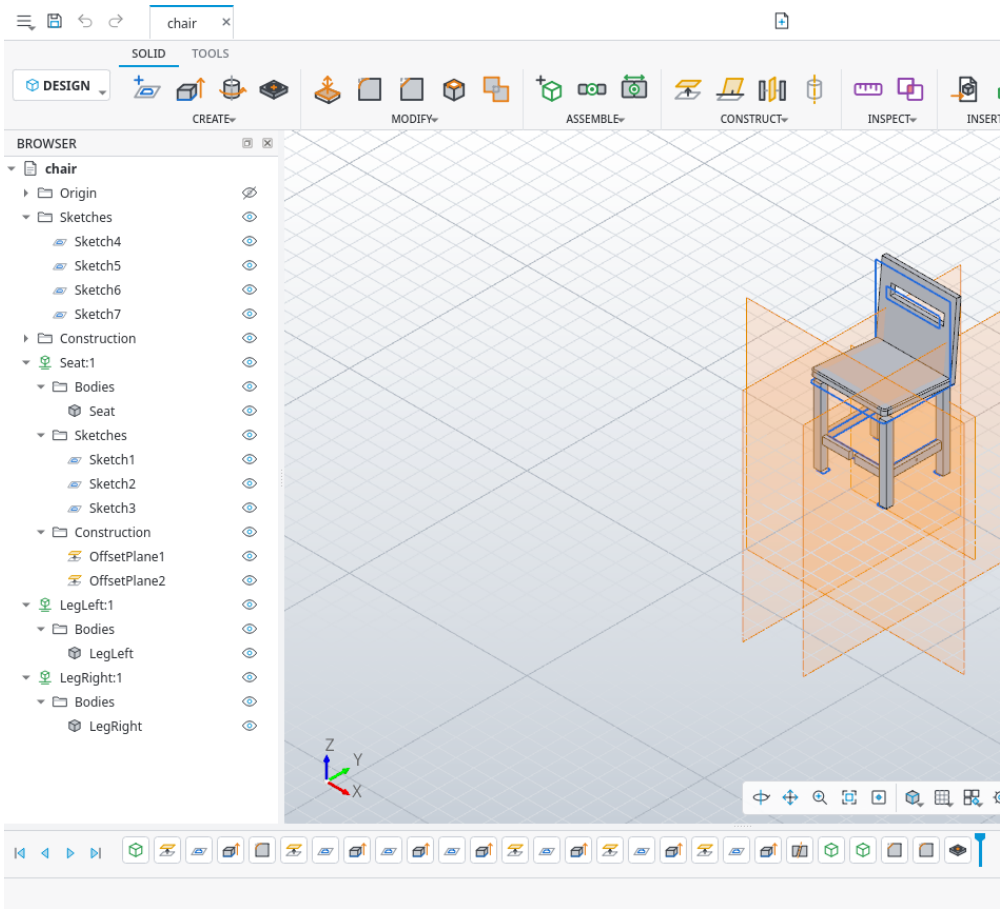
FabriCAD is an open-source parametric CAD application (C++20, Qt 6, OpenCASCADE 7.9) with a Fusion 360–style interface for parts, assemblies and engineering drawings. This report covers the state of the app after the overnight test-and-fix sequence: what was built and fixed, what has been tested, what is still untested, and what is missing from the current scope.

| | | | |
|---|---|---|---|
| **57/57** automated tests passing (Release build) | **129** defects triaged in 3 overnight test rounds | **104** of the 129 verified fixed on retest (+19 partial) | **0** regressions found on retest |
| **~96k** lines of own C++ (+13k vendored PlaneGCS) | **127** commits on main before this report (no remote) | **0.22 s** offscreen GUI startup (FreeCAD 1.1.4: 2.64 s) | **2** issues retested as not fixed |

## At a glance

- **Usable end to end:** sketch > features > assembly with joints > drawing > STEP/STL/PDF/SVG/DXF export. The 3-part chair benchmark builds exactly to spec (B-rep volumes within 0.00002 % of the analytic values) and is a regression test; an 84 s recording of it is at *docs/uiux/chair-run.mp4*.
- **Overnight:** UI/UX, functional/reliability and general-bugs rounds each ran test > triage > plan > fix > merge > retest, with Sonnet testers driving the real GUI using only mouse and keyboard.
- **Not reached:** the dedicated drawing round could not start before the 09:00 cutoff, so the Drawing workspace is the least-tested area.
- **Your reported issues:** light-on-light text, eye toggles needing a redraw and unselectable faces/edges are fixed. The ViewCube now drags to orbit and got larger edge/corner zones, but testers still found it low-contrast, the zones small and corner clicks unreliable.

# 1. Current app state




The chair benchmark open in the current build: browser with components, bodies, sketches and construction planes; timeline with every feature; Fusion-style toolbar.

| Area | State |
|---|---|
| Build | Release build in *build/*, up to date with main; 57/57 ctest entries pass (unit, GUI-logic, end-to-end, chair benchmark, scripting; 16 s with -j2). A clean rebuild shows 6 compiler warnings (4 -Wmaybe-uninitialized on ShapeRef in Selection.cpp, 2 in a GUI test). Run with *build/bin/fabricad*. |
| Interface | Light theme across menus, dialogs, popups; ribbon with SOLID and TOOLS tabs plus a contextual SKETCH tab (SOLID panels CREATE/MODIFY/ASSEMBLE/CONSTRUCT/INSPECT/INSERT/SELECT); browser, timeline, ViewCube, nav bar, S search, cmd> line, marking menu, document tabs, Preferences (units, precision, zoom direction); 313 SVG icons. |
| Modelling | Constraint-driven sketcher (PlaneGCS) with on-screen value boxes; parametric history with rollback, edit, suppress; drag manipulators on features. |
| Assemblies | Components, ground, 7 joint types with drive, as-built joints, move/rotate triad, copy/paste instances, Paste New, component patterns, interference, BOM. |
| Drawings | Sheets and title block, base/projected/section/detail/break views, dimensions, tolerances, GD&T, callouts, parts list and balloons; PDF/SVG/DXF export. |
| Automation | Headless *--script* mode (same command line as the GUI) with *--step-delay* for watchable runs; UI test harness *tools/uitest/ui.sh*. |
| Performance | Offscreen GUI startup ~0.22 s (FreeCAD 2.64 s); chair script 2.6 s headless (FreeCAD 1.3 s): live-preview recomputes per typed property, full recomputes on delete and a 2-3 ms per-command floor. Measured 20 Oct at b246294, not re-measured since. |

# 2. Features added and fixed

| Stage | What was added or fixed |
|---|---|
| Round 1 (build) | Core document/timeline/undo, sketcher + solver, part features (extrude, revolve, sweep, loft, fillet, chamfer, shell, hole, patterns, mirror, combine, press-pull, planes/axes/points), assemblies with joints, drawings, Fusion-style shell, STEP/STL I/O. |
| Round 2 (features) | Icons; sketch value boxes that create dimensions; on-canvas manipulators; document tabs and scripting mode; Split Body, primitives, threads, tapped holes, variable fillet, rib, thicken, scale, Paste New, component patterns; ellipses, splines, sketch text, sketch patterns; drawing tolerances, GD&T, callouts, break views. |
| Chair benchmark | 7 defects fixed: scoped auto-join, fresh STL meshing, holes cut only their own body, safe component delete, IMPORT path, STEP part names, Export STL dialog; pivot rotation. |
| UI/UX round | 41 issues: full light theme and readable text; viewport redraws immediately (eye toggles); picking of faces/edges/bodies restored; ViewCube drag-to-orbit, larger hit zones, hover, home/roll/step arrows; dialogs dock top-left (moved to the right edge in the functional round); fit margins; labelled origin planes. |
| Functional round | 39 issues: per-instance autosave recovery, clear load/feature-failure messages, delete asks about dependents, pattern count capped, over-thick shell handled, command dialogs dock at the right edge, assembly and sketch reliability fixes. |
| General round | 49 issues: closing a design closes its drawings, sheets panel stays in sync with undo, failed save keeps name, suppressed inputs mark dependents as errors, modified flag follows undo, Preferences dialog with units, thread cutter, expression errors block OK, ellipse/spline/text tool fixes. |

## Retest results by area

Counted row by row from the verdict tables in *docs/uiux/retest* and *docs/tests/*/retest*. Partial includes "partially" and "mostly fixed". Not verified: the tester could not reproduce or confirm the case. New: findings as reported, without harness-only notes; most are low, one is medium (rectangle value boxes) and one high (origin planes picked in Joint mode, later fixed as functional ASM-06).

| Round | Area | Fixed | Partial | Not fixed | Not verified | New |
|---|---|---|---|---|---|---|
| UI/UX | Viewer | **13** | 4 | 1 | 0 | 3 |
| UI/UX | Sketch | **15** | 2 | 0 | 0 | 5 |
| UI/UX | Solid | **13** | 0 | 0 | 0 | 6 |
| UI/UX | Assembly + drawing | **5** | 4 | 0 | 0 | 6 |
| Functional | Sketch | **10** | 1 | 0 | 1 | 0 |
| Functional | Part | **7** | 0 | 0 | 0 | 3 |
| Functional | Documents | **7** | 1 | 0 | 0 | 2 |
| Functional | Assembly | **9** | 3 | 0 | 0 | 2 |
| General | Advanced part | **6** | 0 | 0 | 2 | 4 |
| General | Advanced sketch | **13** | 2 | 1 | 0 | 3 |
| General | App tools | **10** | 1 | 0 | 0 | 3 |
| General | Bug hunt | **11** | 3 | 0 | 1 | 2 |
| **Total** | verdict rows | **119** | **21** | **2** | **4** | **39** |
| **Unique** | the 129 triaged issues | **104** | **19** | **2** | **4** | |

Most UI/UX issues were retested in more than one area, so the verdict rows add up to more than the issue count. Per unique issue: UI/UX 32 fixed, 8 partial, 1 not fixed; functional 33, 5, 0 (1 not verified); general 39, 6, 1 (3 not verified).

# 3. Features tested

All overnight testing was done through the real GUI with mouse and keyboard only, on private X displays, and observed through screenshots. Earlier rounds also added automated tests and a scripted chair benchmark.

| Area | Tested and working | Result |
|---|---|---|
| Sketch | Lines, rectangles, circles, arcs, polygons, slots, ellipses, splines, text, patterns, move/rotate/scale, trim/extend/break/offset/fillet/chamfer, project, constraints, dimensions, edge cases (zero-size, 0.01 mm-10 m, conflicts), sketch on angled faces | Good |
| Part features | Extrude (new body/join/cut/intersect; one side, two sides, symmetric, all; not To Object or Thin), revolve, fillet (constant/variable), chamfer, shell, hole (simple/counterbore/tapped), patterns, mirror, combine, press-pull, primitives, coil, threads, draft, sweep, loft, rib, thicken, scale, split, planes/axes/points | Good |
| History | Timeline edit, rollback, insert, suppress, delete with dependents, long undo/redo | Good |
| Documents | Save/open fidelity (20-30 feature models, 17 sketches), Save As, tabs, autosave recovery after kill -9, corrupt files, STEP/STL export and import | Good |
| Assemblies | Components, ground, rigid/revolute/slider joints, drive, move/rotate triad, copy/paste, nested components, split-then-components | Mostly good |
| App tools | Measure, interference, materials, appearance, parameters with units, S search, cmd> line, marking menu, shortcuts, help, context menus | Good |
| Viewer / UI | Navigation, ViewCube, display modes, selection and filters, visibility, theme and contrast | Mostly good |
| Benchmark | 3-part chair vs FreeCAD 1.1.4: exact geometry, watertight STL, STEP part names, timing | Passes |
| Drawings | Only touched incidentally: sheet and title block, one base/projected/section view, one linear dimension, parts list, PDF export, sheet add/delete/undo | Thin |

## Still open after retest

- ViewCube is still faint on a light background, and a click near a corner sometimes gives an edge view (UX-05).
- Origin child items have no eye toggles (UX-40). Origin planes can be drawn oversized and look doubled (UX-10).
- After Finish Sketch no profile shading is shown (general-round SK-12, not fixed).
- Selection: no freeform or paint selection (box window/crossing select works), and no check mark on the active priority or filter: the check state is defined but the ribbon never shows it (UX-39).
- Rectangle value boxes can sit under the cursor and swallow the second-corner click (new, medium).
- The extrude value box does not follow the handle after an orbit (new, low).
- Only partly fixed: Open Recent entries for same-named files still collide (functional DOC-11); overlapping pattern bodies get a warning but OK stays enabled (general PART-10).

# 4. Features not yet tested

- **Drawing workspace in depth:** detail and break views, and projected/section views beyond one simple case; annotation types other than a linear dimension; associativity after model edits; balloons; quality of the exported SVG/DXF files (only a PDF export was checked). The dedicated drawing round is prepared and ready to run.
- **HiDPI and scaling:** nothing was tested at 2× device pixel ratio, and the ViewCube size is in device pixels.
- **Native Wayland:** the system OpenCASCADE is X11/GLX only, so all testing ran through XWayland.
- **Scale:** large assemblies (hundreds of parts) and very long histories, beyond the 20-35 feature test models.
- **Round-trips with other CAD:** STEP files from other CAD systems, and files from FreeCAD or Fusion.
- **Live GUI timing:** click and keystroke counts versus FreeCAD were estimated from code, not measured.

# 5. Missing features (within the current scope)

The scope is a fast, lightweight, Fusion-like parametric modeller for parts, assemblies and drawings. These gaps were found by testers, in reports, or by checking the source.

| Area | Missing or incomplete |
|---|---|
| Application | Preferences cover only units, display precision and zoom direction (no theme or display options); dark theme; HiDPI tuning (ViewCube is sized in device pixels); Scripts entry under TOOLS; separate headless runner without the main window. |
| Sketch | Sketch-level profile shading after Finish Sketch; offset dimension for curves; constraint for point-to-curve distance; absolute coordinates while drawing create no constraints. |
| Part | Sheet metal; surface-modelling workspace; boundary fill; section analysis and zebra inspection; manipulators for sweep, loft, draft, patterns, thread and split. |
| Assembly | Motion links, contact sets and motion study; exploded views; assembly structure kept on STEP import (it is flattened into bodies). |
| Drawing | Exploded and animated views; DXF import into sketches; depth of standards compliance not verified. |
| Selection / UI | Freeform and paint selection (box window/crossing select exists); check marks on the active selection priority/filter; eye toggles on Origin children; ViewCube contrast. |
| Performance | Scripted builds are about 2× slower than FreeCAD (preview recomputes per typed value, full recompute on delete); the default STL tessellation is coarser than FreeCAD's. |

## 6. Where things are

| Item | Location (in ~/Documents/projects/fabricad) |
|---|---|
| App binary | build/bin/fabricad (--script, --step-delay, --offscreen) |
| Test reports | docs/uiux/, docs/tests/functional/, docs/tests/general/ (round1, plans, fixes, retest) |
| Benchmark | docs/comparison/COMPARISON.md, CHAIR_SPEC.md, tests/chair/ |
| Recording | docs/uiux/chair-run.mp4 (84 s) |
| Next step | Run the drawing round (args ready in the session scratchpad test-plan.json) |